

Submit your Jupyter notebook with the answers to gradescope by 3:40pm.

1. Plot the cylinder $(x - 2)^2 + (y - 2)^2 = 1$ (blue) and the plane $z = 1 + x + y$ (green), selecting a good range for the variables.

Draw the curve in the intersection of the cylinder and the plane in red.

2. Consider the polynomial system

$$P = \{x^2 - 3xyz + 4y - 3 = 0, yz + x - 5 = 0, z^2 + x - y + 1 = 0\}.$$

- (a) Convert P into a triangular form.
- (b) How many solutions does the system have? Justify your answer.
- (c) Verify your prediction on the number of solution by solving the system.